

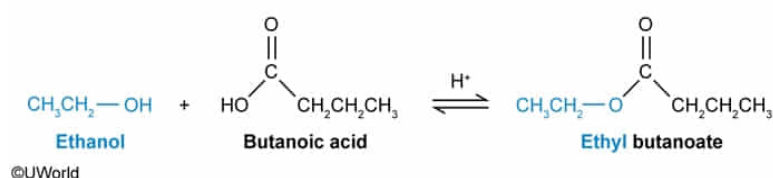
What is the product of the reaction of butanoic acid and ethanol under acidic conditions?

- ☐ A. Butyl ethanoate [19%]
- ✓ ☒ B. Ethyl butanoate [65%]
- ☐ C. Butanoic ethanoic anhydride [9%]
- ☐ D. Acetic butanoic anhydride [5%]

Correct

65% answered correctly

### Explanation:



**Esters** are **carboxylic acid derivatives** formed by a **reaction** between a **carboxylic acid** and an **alcohol**. Under acidic conditions, this reaction is known as a **Fischer esterification**. The first step of Fischer esterification is protonation of the carbonyl oxygen, followed by nucleophilic attack of the electrophilic carbonyl carbon by the alcohol. Next, a proton is transferred from the alcohol oxygen to the hydroxyl group of the carboxylic acid, resulting in the formation of water (a **good leaving group**). Finally, the carbonyl oxygen is deprotonated to yield an ester.

For esters, the alkyl group (from the alcohol) is named first as the substituent prefix, followed by the carboxylic acid. The **suffix** *-ic acid* is then replaced by **-ate**. Therefore, the reaction of butanoic acid and ethanol by a Fischer esterification yields **ethyl butanoate**.

**(Choice A)** **Butyl ethanoate** is an ester made from ethanoic acid

and butanol.

**(Choices C and D)** The reaction forms an ester rather than an **anhydride**. Anhydrides are made most commonly from a **carboxylic acid and an acid chloride**, and are named by stating the two carboxylic acids alphabetically and replacing *acid* with *anhydride*; butanoic ethanoic anhydride and acetic butanoic anhydride refer to the same compound because "acetic" is the common term for "ethanoic."

**Educational objective:**

Esters are carboxylic acid derivatives and may be formed by the condensation of a carboxylic acid and an alcohol under acidic conditions (Fischer esterification). Esters are named by using the alkyl group from the alcohol as a substituent prefix and the carboxylic acid as the root name, with the *–ic acid* suffix replaced by *–ate*.

**Time spent:**0 sec

**Subject:**

Organic Chemistry

**QID:**401558

**System:**

Functional Groups and Their Reactions